

Integrating Vision and Multimodal Language Models for Collaborative Building Footprint Quality Assessment

Extracting building footprints has long been an essential task in aerial imagery interpretation and GIScience, supporting applications such as disaster management, urban planning, and infrastructure monitoring (Awrangjeb et al., 2010). Accurate and up-to-date building footprints are not only required by national mapping agencies for maintaining topographic databases, but also play a critical role in crisis situations (Zheng et al., 2021) for example, after hurricanes in Jamaica, earthquakes in Afghanistan, or the war in Ukraine (Aimati, 2023). In such contexts, quickly assessing structural damage and urban change is vital for response planning and reconstruction. Despite significant advances in deep learning and remote sensing automation, mapping authorities such as swisstopo still rely heavily on manual verification and editing. According to a short informal interview with an operator, there is currently no fully operational automated system capable of reliably generating building footprints for new or modified buildings in their production workflows.

The proposed workflow will be developed and tested in different stages. For the development of the method, I will first start with building footprint extraction in Switzerland, since here I have access to very good satellite and aerial imagery as well as excellent building footprint data that can serve as a benchmark. Once the approach is established and performs well, I will then choose a specific use case to apply the model for example, a natural disaster such as a flood or earthquake, or an armed conflict area where large-scale building damage has occurred. This use case still needs to be selected, but it will help to show how the method performs in more challenging real-world GIS situations

In recent years, with the rise of artificial intelligence, researchers have increasingly explored the use of foundation models instead of traditional deep learning architectures. Most deep learning models (including CNNs) rely on large, labeled datasets and often generalize poorly to new domains. The Vision Transformer (ViT) (Dosovitskiy et al., 2020) was one of the first transformer-based models for vision tasks, demonstrating that attention mechanisms can effectively replace convolutional operations. Building on this idea, the Dense Prediction Transformer (DPT) (Ranftl et al., 2021) further showed that transformers can produce coherent, global predictions for dense visual tasks. Recently, research has moved toward integrating visual foundation models with multimodal large language models (MLLMs) (Wu & Gan, 2023). These models can process visual and textual information, offering new ways to interpret and reason about spatial data (Ji et al., 2025). In GIScience, such approaches could be crucial for enhancing spatial reasoning and enabling automated analysis of geospatial features beyond pixel-level classification. Early studies have begun to explore these combinations (Zhou et al., 2024), typically by allowing the language model to guide or describe the visual model's segmentation process.

My project takes a different approach. Instead of using a language model to directly control a vision model, I propose a two-step framework that separates generation and evaluation. In

the first stage, rough building polygons are generated using a visual foundation model. In the second stage, a multimodal language model (MLLM) evaluates these polygons, identifies geometric errors, and suggests improvements or describes them. This two-step design is valuable because it addresses key limitations of current one-step or end-to-end approaches, which typically require large labeled datasets and are difficult to adapt across regions or data domains. By decoupling generation, evaluation and correction the proposed workflow allows for clearer feedback loops and greater flexibility in integrating automated quality assurance into geospatial production processes. The research gap lies in the limited transfer of such multimodal, feedback driven AI approaches to GIScience, particularly in the automated quality assessment of vector based building footprints.

Rough Research questions:

- RQ1: How accurately can multimodal large language models (MLLMs) detect and describe typical geometric errors in building polygons?
- RQ2: How effective is the conversion of MLLM generated feedback into spatial refinements for improving the geometric quality of building polygons?
- RQ3: How well does the proposed workflow generalize and perform when applied to a real-world use case (such as updating building footprints after natural disasters or in conflict-affected areas)?

I tested this approach using two foundation models. As the multimodal large language model (MLLM), I used the OpenAI API with GPT-5, and as the visual model, I used the *Segment Anything Model* (SAM) developed by Meta AI (Kirillov et al., 2023). Both models are pretrained on a large amount of data (Bommasani et al., 2021) and therefore do not necessarily require additional fine tuning.

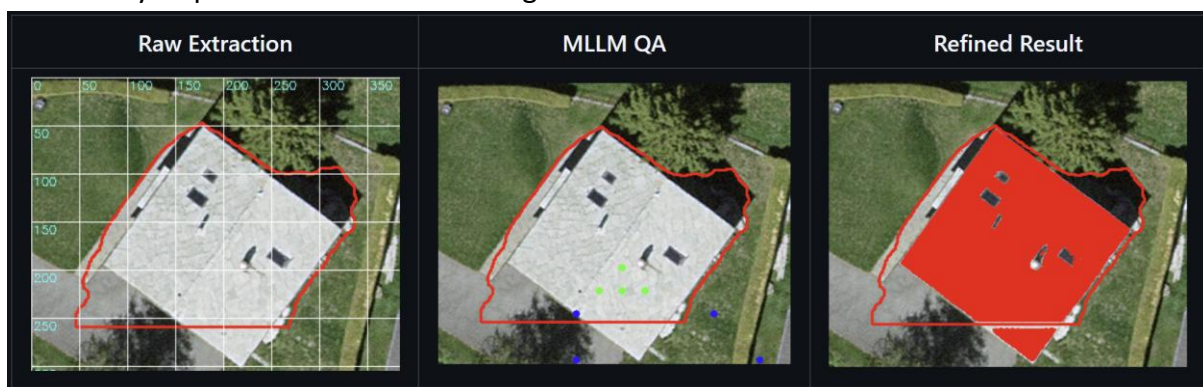


Figure 1 Overview of the tested workflow. Raw Extraction shows the initially generated building polygon with an overlaid grid used for MLLM processing. MLLM QA illustrates the quality assessment step, where the multimodal language model identifies and suggests geometric improvements. Refined Result displays the final building footprint after applying the suggested refinements (in solid red).

For the final master's thesis, it still needs to be decided which foundation models such as SAM, GPT, LLaVA 1.6 / 1.7 (Vicuna-13B), or InternVL 2 (8B) will be used.

As a reference and benchmark, either the swisstopo TLM dataset together with SwissImage aerial imagery ([swisstopo, 2025](#)) or publicly available building datasets such as Microsoft

Open Buildings and Google Open Buildings will be used.

The swisstopo dataset is subject to licensing agreements that restrict its use with commercial services such as the OpenAI API ([Terms of Use](#)). Therefore, the goal is to use an open source multimodal model that can be run locally in a Docker container or on a Swiss-based server to ensure compliance with data protection and licensing requirements.

Period	Phase	Key Tasks & Milestones
Nov – Dec 2025	Literature Review & Concept Development	- Read and summarize core papers (CNNs → DPT → SAM → MLLM). - Define research questions, objectives, and hypothesis. - Write Introduction draft and Theoretical framework. - Set up environment (Docker, GPU access, Swisstopo data). <i>Milestone 1: Approved proposal & first theory draft.</i>
Jan – Feb 2026	Visual Model Implementation (eg SAM)	- Implement footprint extraction workflow. - Evaluate baseline results (IoU, precision, recall). - Continue writing Methodology – Part 1 (Visual Model). <i>Milestone 2: Working baseline & documented visual workflow.</i>
Mar 2026	MLLM QA Integration	- Implement and test MLLM evaluation stage (GPT-4V, InternVL2, etc.). - Define JSON schema for issues & suggestions. - Draft Methodology – Part 2 (MLLM QA). <i>Milestone 3: Functional QA pipeline with description in text.</i>
Apr – May 2026	Model Coupling & Framework Development	- Integrate MLLM + Visual Model into single pipeline. - Automate improvement suggestions and feedback loop. - Update Methodology with diagrams and flowcharts. <i>Milestone 4: Prototype framework complete & method fully written.</i>
Jun 2026	Evaluation & Analysis	- Expand Application case (other countries, change detection etc) - Compare performance vs. baseline and between models. - Begin Results chapter (quantitative tables, figures). <i>Milestone 5: Evaluation dataset processed & first results written.</i>
Jul – Aug 2026	Discussion, Conclusion & Finalization	- Write Discussion, Conclusion, and Abstract. - Final proofreading, formatting, and supervisor feedback. - Prepare presentation (if required). <i>Milestone 6: Final thesis submission (mid-August).</i>

AI Statement: ChatGPT 5 was used to help generate and structure the timeline table, and to refine the wording. It also supported me in improving sentence structure, understanding, spelling, and grammar. In addition, it assisted with citation formatting.

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